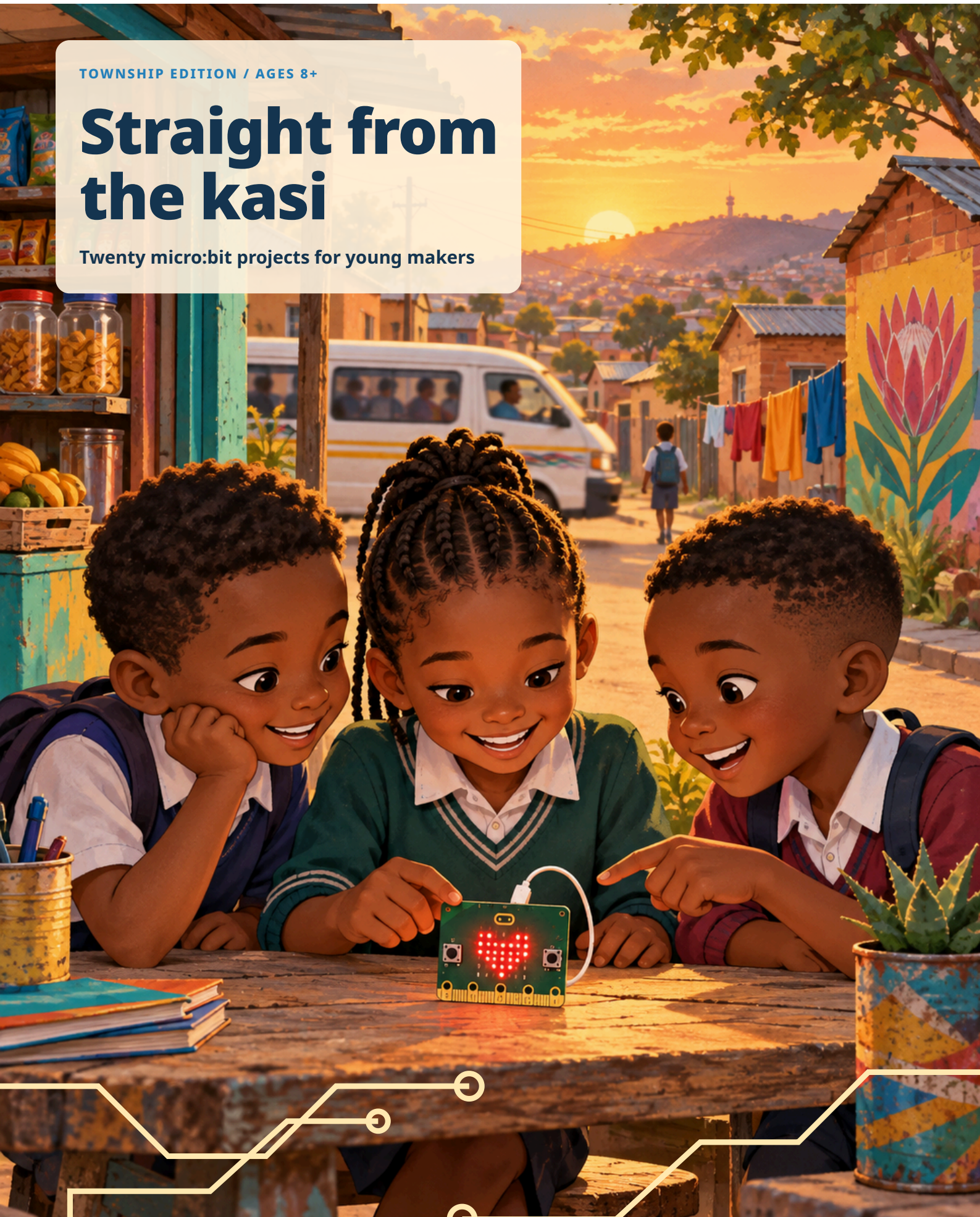


TOWNSHIP EDITION / AGES 8+

Straight from the kasi

Twenty micro:bit projects for young makers



Every language is a doorway into computational thinking.

Your ideas belong here.

How to use this book

Twenty projects take you from your first flashing heart to your own invention for the kasi.



Illustrative township learning scene

A small computer. A world of possibilities.

The BBC micro:bit is a computer small enough to fit in your hand. It has a 5 x 5 LED display, two buttons, sensors that feel movement, light, temperature and direction, a radio for talking to other boards - and on the V2 board, a speaker, a microphone and a gold touch logo. You code it with Microsoft MakeCode: coloured blocks that snap together like puzzle pieces.

IDEAS

CODE

CREATION

COMMUNITY

OUR CIRCUIT KIDS SIGNATURE

**"Every language is a doorway
into computational thinking."**

This book belongs to _____

Follow the story. Make it yours.

Projects 1 to 6 build your confidence. Projects 7 to 12 explore sensors and data. Projects 13 and 14 connect boards over radio. Projects 15 to 19 add wires, lights, sound and movement. Project 20 is all you: an invention for your own community.

1-6

Confidence

7-12

Sensors + data

13-14

Radio

15-19

Build + connect

20

Your invention

01 / The story

Every project opens in the kasi, with a problem or a moment you already know.

02 / Build it

Numbered steps name the real MakeCode blocks. Follow them in order - that is what code is.

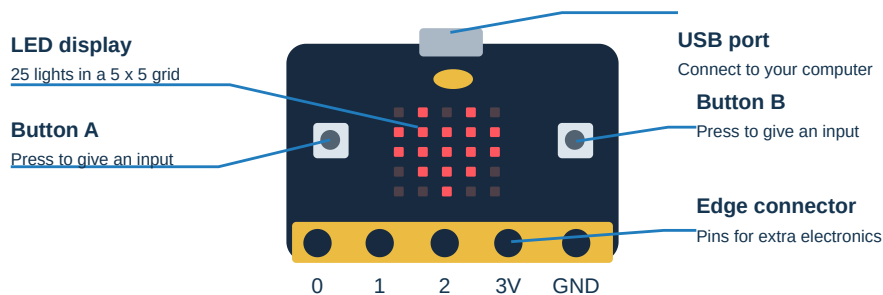
03 / Make it yours

The art and design layer. A project is not finished until it looks like yours.

04 / What you learned

One short paragraph tying the build back to computational thinking.

Meet the front of your micro:bit V2



Original simplified diagram. The gold logo above the LEDs is a touch input.

Learning objectives

- ☐ Follow a sequence of steps in order.
- ☐ Make each project your own through art and design.
- ☐ Connect your build to computational thinking.

One book, three editions

This is the Township edition. The Rural and Christian editions carry the same twenty projects with stories re-rooted in each community's life - every learner meets the same skills at their own doorway.

Small kit. Big ideas.

Your tools and safety, every time

Almost everything in this book runs on the micro:bit GO kit alone. Four projects (15, 16, 18 and 19) ask for extras from the club's extension tray - cheap, reusable and locally sourced.

The GO kit

BBC micro:bit V2 board, USB data cable, battery holder with two AAA batteries. That is the whole kit for sixteen of the twenty projects.



micro:bit V2



USB data cable



Battery holder + 2 AAA

The extension tray

Crocodile leads, three LEDs with resistors, one micro servo, kitchen foil, recycled card and craft materials. Shared by the club.

Your exercise book

For tallies, tables, sketches and designs. The paper is part of the lab.

Safety, every time

Battery power only - never mains electricity. Power off before changing any wiring. The servo is powered from the battery pack, never the board's 3V pin. A facilitator or adult supervises batteries, scissors and small parts. Ages 8+ with adult or facilitator support.



No wifi? No problem.

Coding without internet

This whole book works offline.

- 1 The club laptops run the MakeCode offline app, installed once before any workshop. The browser editor also keeps working after it has loaded once.
- 2 Write your code, click Download, and drag the .hex file onto the MICROBIT drive - the yellow light flashes while it copies.
- 3 The club USB carries tested .hex files for every project. If a laptop fails, the facilitator can flash a working program straight onto your board.
- 4 No accounts, no email addresses, no sign-ups. Your code stays yours.

WRITE CODE

DOWNLOAD

MICROBIT

.hex

Your code stays yours

Code. Copy. Create.

Inside this book

Twenty projects, straight from the kasi

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Sawubona, micro:bit!

Your story. Your code. Your community.

Gogo says the future fits in the palm of your hand. She is right. Today you meet the small computer you will code in every project in this book.

Learning objective

Name every part of the micro:bit, connect it to a computer and run the program it arrives with.

What you need

micro:bit GO kit (board and USB cable), one computer

Blocks you will use

None yet - today is for exploring

Build it / Step by step

- 1 Hold the board with the lights facing you. Find the 5 x 5 LED display and buttons A and B.
- 2 Find the gold touch logo on the front. Turn the board over to find the speaker. Find the microphone, the USB port, the battery connector, and the gold pins labelled 0, 1, 2, 3V and GND.

LED display

25 lights in a 5 x 5 grid

Button A

Press to give an input

USB port

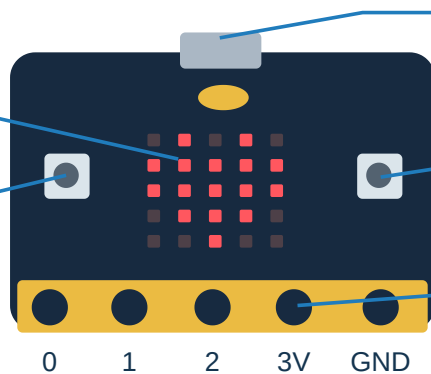
Connect to your computer

Button B

Press to give an input

Edge connector

Pins for extra electronics



Original simplified front schematic

Make it yours.

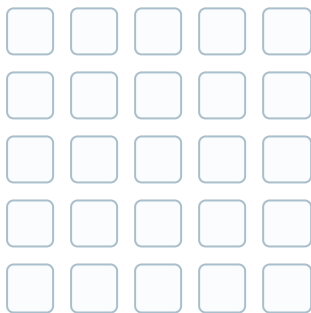
Sawubona, micro:bit!

- 3 Plug the USB cable into the board and the computer. The board wakes up and plays its own demo program.
- 4 Press button A, then button B. Tilt the board. Watch the display and write down what changes.
- 5 Draw your board in your exercise book and label five parts - in your home language first, then in English.

Make it yours / Challenge

Design a name sticker for your board or battery pack. Every maker in the kasi marks their tools - mark yours.

PLAN YOUR 5 X 5 DESIGN



INPUT



PROCESSING



OUTPUT

What you learned

A computer is a system. Inputs (buttons, sensors) go in, your code does the processing, and outputs (lights, sound) come out.

My idea / what I would improve

Heartbeat of the kasi

Your story. Your code. Your community.

Stand outside at night and listen: taxis humming, drums somewhere far, gogo's feet on the kitchen floor. Everything alive has a rhythm. Your first program gives your micro:bit a heartbeat.

Learning objective

Code a heart that beats on the LED display, over and over, forever.

What you need

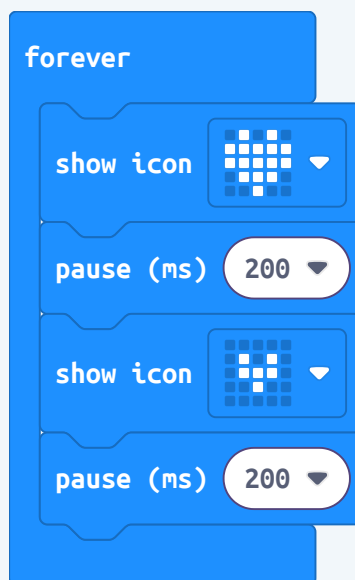
GO kit, one computer

Blocks you will use

"forever", "show icon", "pause (ms)"

Build it / Step by step

- 1 Open MakeCode and start a New Project called heartbeat.
- 2 Drag "show icon" into "forever" and choose the big Heart.



Real MakeCode blocks / example from these steps

Make it yours.

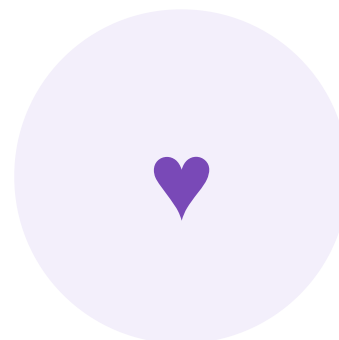
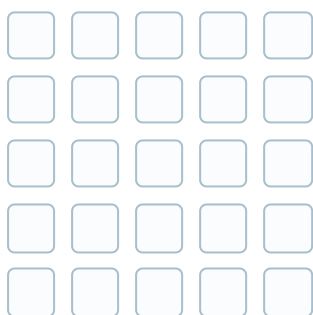
Heartbeat of the kasi

- 3 Add "pause (ms)" 200 under it, then another "show icon" - this time the small heart - then one more "pause (ms)" 200.
- 4 Click Download, then drag the .hex file onto the MICROBIT drive. The yellow light flashes while it copies.
- 5 Watch it beat. Now change both pauses to 500. What changed? Which heartbeat feels like yours after soccer?

Make it yours / Challenge

Open "show leds" and draw your own symbol dot by dot - a star for your street, a taxi, a protea. Swap it in for the small heart.

PLAN YOUR 5 X 5 DESIGN



What you learned

A program is a sequence of steps in order. A loop repeats those steps, and timing is part of the design.

My idea / what I would improve

Name in lights

Your story. Your code. Your community.

Every spaza shop has a painted sign so the whole street knows its name. Time to put your own name up in lights.

Learning objective

A badge that scrolls your name across the display, then flashes your icon.

What you need

GO kit, one computer

Blocks you will use

"forever", "show string", "show icon", "pause (ms)"

Build it / Step by step

- 1 Start a New Project called my-name.
- 2 Inside "forever", place "show string" and type your name. Short names scroll best.



Real MakeCode blocks / example from these steps

Make it yours.

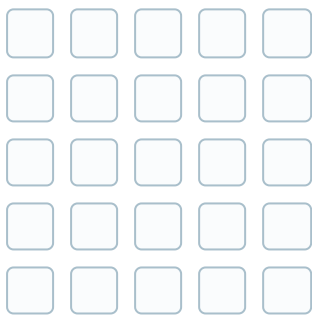
Name in lights

- 3 Add "show icon" after it and pick an icon that feels like you.
- 4 Add "pause (ms)" 500 at the end of the loop so the icon rests before the name comes back.
- 5 Download it, plug in the battery pack, and wear it to the club. You are walking signage.

Make it yours / Challenge

Design a logo for your street-soccer team or your family on 5 x 5 grid paper, then code it with "show leds". Every great brand in the kasi started as one small drawing.

PLAN YOUR 5 X 5 DESIGN



What you learned

Sequences tell a story step by step, and communication is a design choice: a short, clear message always beats a long one.

My idea / what I would improve

Taxi counter

Your story. Your code. Your community.

Stand on the corner at four o'clock. How many taxis pass in ten minutes? Everybody guesses. You are going to count.

Learning objective

A counter: button A adds one, button B resets to zero, shake shows the total.

What you need

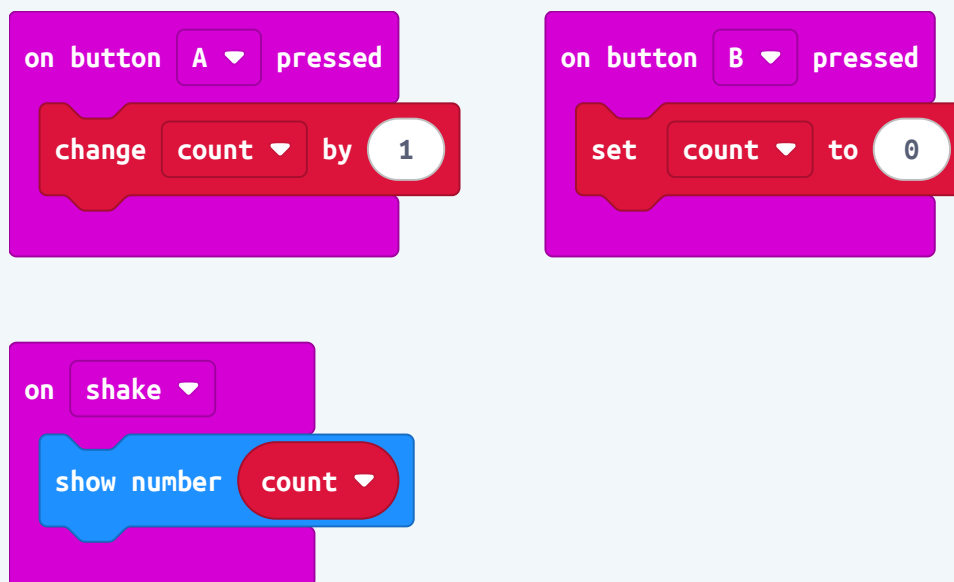
GO kit, one computer

Blocks you will use

"on start", "on button A pressed", "on button B pressed", "on shake", "set count to", "change count by", "show number"

Build it / Step by step

- 1 Start a New Project called taxi-counter.
- 2 Make a variable called count. In "on start", place "set count to" 0.



Real MakeCode blocks / example from these steps

Make it yours.

Taxi counter

- 3 In "on button A pressed", place "change count by" 1. Every press adds one taxi.
- 4 In "on shake", place "show number" count, so you can read the total without resetting.
- 5 In "on button B pressed", place "set count to" 0 for the next count.
- 6 Download it, take it outside, and count something real for ten minutes: taxis, bicycles, gogos with trolleys. Write the number down.

Make it yours / Challenge

Draw a tally poster for your counting station, kasi style, with big painted numbers. Or use your counter as the scoreboard at your next street soccer game.

MY TEST RECORD

Test / place	What happened?



123

What you learned

A variable is a box with a name that stores a changing value, and events like button presses are what make a program respond.

My idea / what I would improve

Kasi dice

Your story. Your code. Your community.

Load-shedding again. No TV, no data, and gogo's old board game lost its dice under the couch years ago. Make a dice that can never get lost.

Learning objective

Shake the board to roll a number from 1 to 6, like a fair dice.

What you need

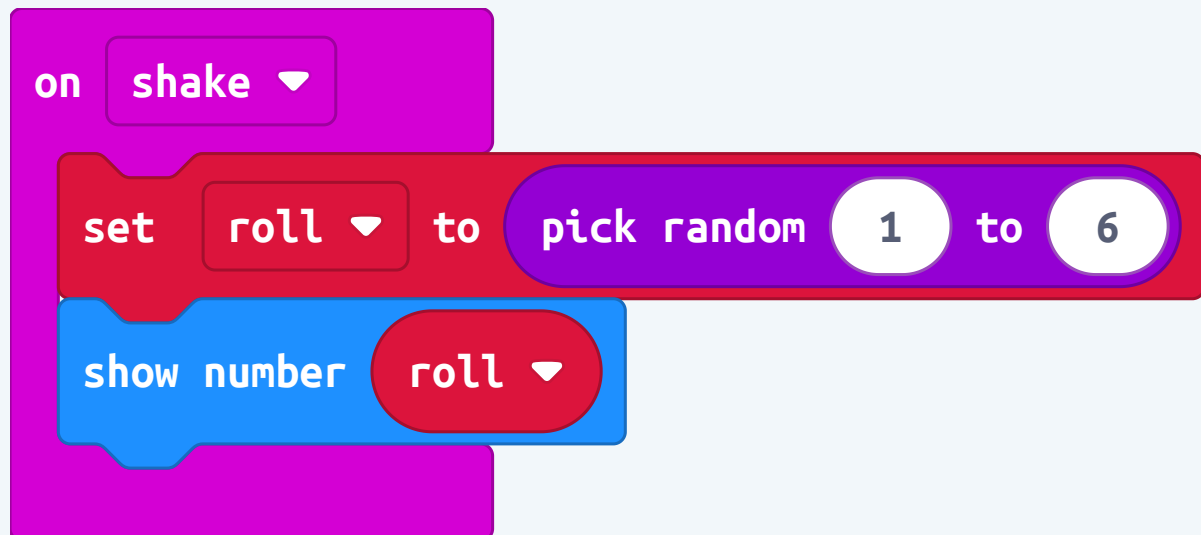
GO kit, one computer

Blocks you will use

"on shake", "set roll to", "pick random 1 to 6", "show number"

Build it / Step by step

- 1 Start a New Project called dice and make a variable called roll.
- 2 In "on shake", place "set roll to" and snap "pick random 1 to 6" into it.



Real MakeCode blocks / example from these steps

Make it yours.

Kasi dice

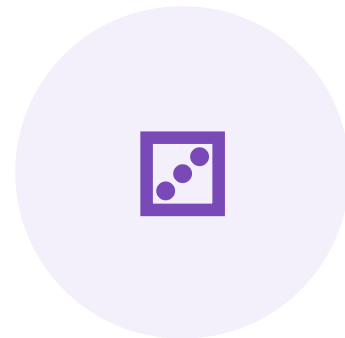
- 3 Under it, place "show number" roll.
- 4 Download it. Shake, read, play. The accelerometer inside the board feels your shake.
- 5 Test if it is fair: shake 30 times and tally each number. Did every number get a chance?

Make it yours / Challenge

Design the board game your dice belongs to: snakes and ladders, but the ladders are taxis going up and the snakes are potholes. Draw it on flattened cardboard.

MY TEST RECORD

Test / place	What happened?



What you learned

Randomness makes games fair, sensors turn movement into input, and a good tester checks the numbers instead of trusting them.

My idea / what I would improve

Rock, paper, scissors

Your story. Your code. Your community.

Who buys the amagwinya? Who bats first? The corner has always settled arguments with rock-paper-scissors. Now your board can play too.

Learning objective

Shake the board and it randomly shows rock, paper or scissors.

What you need

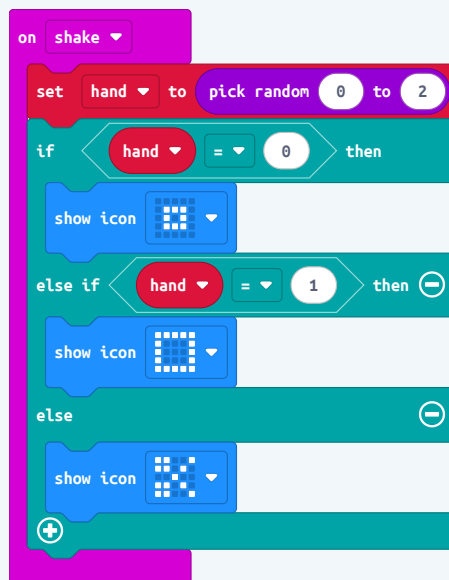
GO kit, one computer

Blocks you will use

"on shake", "set hand to", "pick random 0 to 2", "if ... then ... else", "show icon"

Build it / Step by step

- 1 Start a New Project and make a variable called hand.
- 2 In "on shake", place "set hand to" with "pick random 0 to 2" inside it.



Real MakeCode blocks / example from these steps

Make it yours.

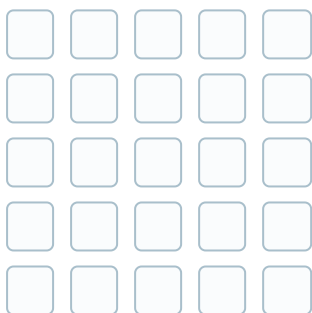
Rock, paper, scissors

- 3 Add "if hand = 0 then" and inside it "show icon" Small Square - that is the rock.
- 4 Add "else if hand = 1" with "show icon" Square - that is the paper.
- 5 In the last "else", place "show icon" Scissors.
- 6 Download and play a friend. First to five wins, loser buys the amagwinya.

Make it yours / Challenge

Invent your own three symbols on grid paper and code them with "show leds". Taxi beats bicycle, bicycle beats skateboard? Your game, your rules.

PLAN YOUR 5 X 5 DESIGN



What you learned

Selection - if, then, else - lets a program choose between paths, and clear symbols are a design decision your player must understand.

My idea / what I would improve

How are you today?

Your story. Your code. Your community.

At the club we check in before we code. Gogo asks it at the gate every afternoon: how are you today? Make a badge that answers for you.

Learning objective

Press A for happy, B for sad, A+B together for surprised - and on a V2 board, touch the gold logo for sleepy.

What you need

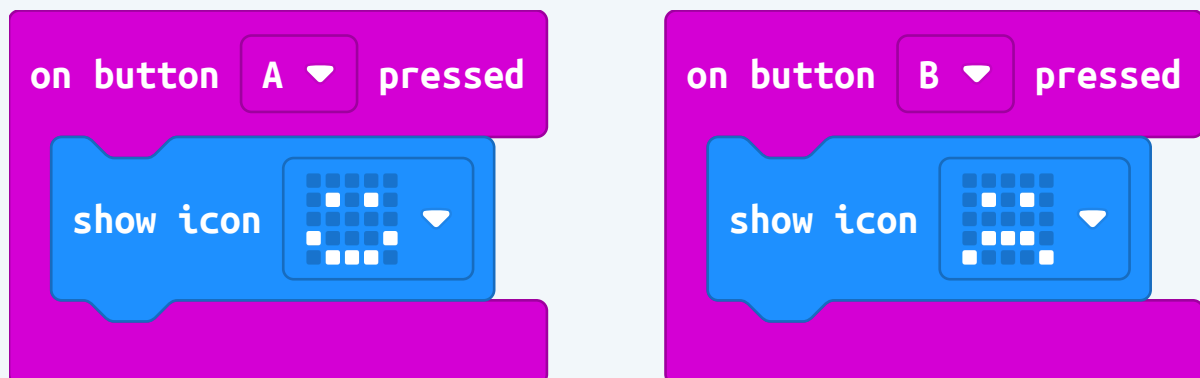
GO kit, one computer

Blocks you will use

"on button A pressed", "on button B pressed", "on button A+B pressed", "on logo pressed", "show icon"

Build it / Step by step

- 1 In "on button A pressed", place "show icon" Happy.
- 2 In "on button B pressed", place "show icon" Sad.



Real MakeCode blocks / example from these steps

Make it yours.

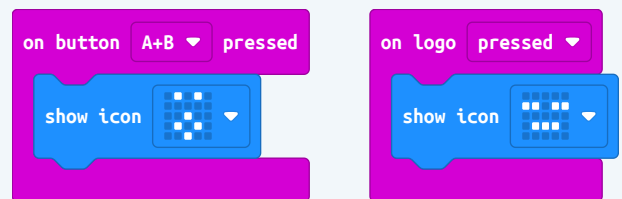
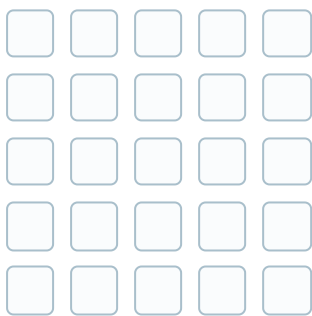
How are you today?

- 3 In "on button A+B pressed", place "show icon" Surprised. Two buttons together is a new input of its own.
- 4 If your board is a V2, add "on logo pressed" with "show icon" Asleep for the days after a long week.
- 5 Download it and check in at the club. Let your badge talk first.

Make it yours / Challenge

Design two brand-new emotion faces on 5 x 5 grid paper - proud, worried, bored, hungry - and code them with "show leds". Naming a feeling is art and science at once.

PLAN YOUR 5 X 5 DESIGN



Real MakeCode blocks / second event

What you learned

Events make a program interactive, combining inputs (A+B) multiplies what two buttons can say, and boolean logic - AND - is hiding inside A+B.

My idea / what I would improve

Load-shedding light meter

Your story. Your code. Your community.

Stage 4 again and the lights are out at six. Where is the best spot to finish homework before the candles? Your board can measure light - its LED display works as a light sensor.

Learning objective

A live light meter that draws a bar graph of the light level, and shows the exact number when you press A.

What you need

GO kit (battery pack for walking around), one computer

Blocks you will use

"forever", "on button A pressed", "plot bar graph of light level up to 255", "light level", "show number"

Build it / Step by step

- 1 In "forever", place "plot bar graph of light level up to 255".
- 2 In "on button A pressed", place "show number" with "light level" inside it.



Real MakeCode blocks / example from these steps

Make it yours.

Load-shedding light meter

- 3 Download and unplug: walk around your home on battery power and watch the bars rise and fall.
- 4 Measure five places - by the window, the kitchen, the darkest corner - and write each number down.
- 5 Mark the brightest spot. That is the load-shedding homework corner.

Make it yours / Challenge

Draw a light map of your home: colour the bright places yellow, the dark places blue, and put a star where the homework corner is. Pin it on the fridge so the whole family knows.

MY TEST RECORD

Test / place	What happened?



What you learned

Sensors turn the world into data, and data beats guessing. Measuring is how you make a decision you can defend.

My idea / what I would improve

The coolest corner

Your story. Your code. Your community.

Soweto summer, 32 degrees before lunch. Gogo keeps the amasi in the coolest place in the house - she just knows. Can you find it with science?

Learning objective

A thermometer: press A to show the temperature, then run a fair test around your home.

What you need

GO kit, one computer

Blocks you will use

"on button A pressed", "temperature (°C)", "show number"

Build it / Step by step

- 1 In "on button A pressed", place "show number" with "temperature (°C)" inside it.
- 2 Choose four places to test: a sunny window, a shady floor, the fridge door, gogo's cool spot.

on button **A ▼** pressed

show number temperature (°C)

Real MakeCode blocks / example from these steps

Make it yours.

The coolest corner

- 3 Test fairly: put the board down - your warm hand changes the reading - wait one minute, then press A.
- 4 Record all four numbers in a table in your exercise book.
- 5 Report to the house: here is the coolest corner, and here is the proof.

Make it yours / Challenge

Draw a heat map of your home or classroom, from hot red to cool blue. Tape it up and see if the family agrees with your data.

MY TEST RECORD

Test / place	What happened?



What you learned

A fair test changes one thing at a time and keeps everything else the same. That is how data becomes evidence instead of opinion.

My idea / what I would improve

Backpack guard

Your story. Your code. Your community.

The taxi rank is full and your bag carries your whole life - books, lunch, the soccer socks. Build a guard that shouts the moment anyone moves it.

Learning objective

A motion alarm: when the board is shaken, it flashes an angry face and plays three short tones. Press A to clear the display and stop any sound still playing.

What you need

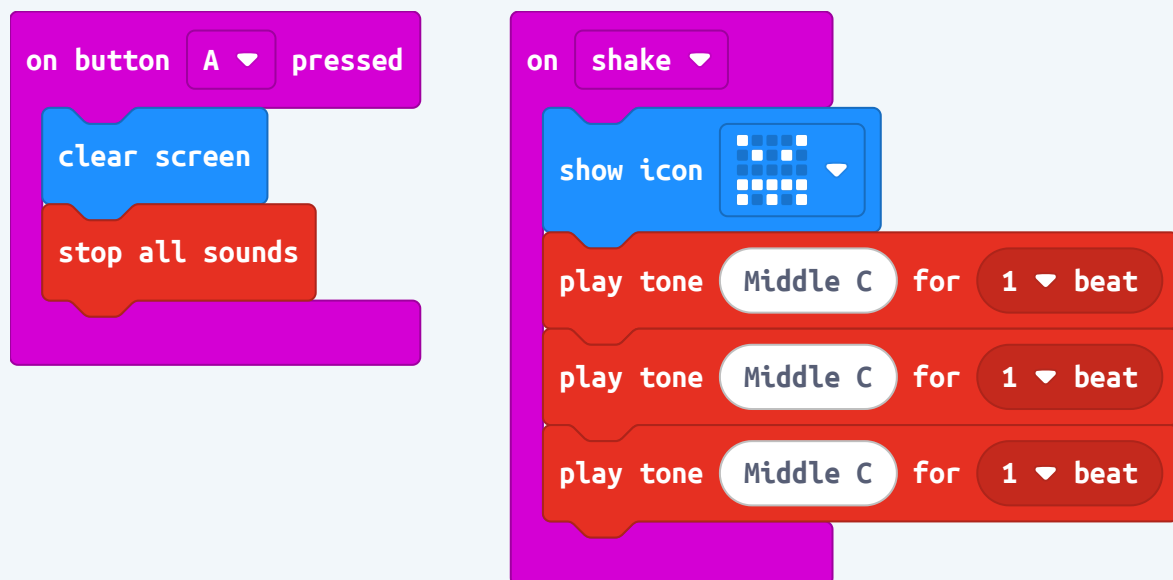
GO kit (V2 board for sound), one computer

Blocks you will use

"on shake", "on button A pressed", "show icon", "play tone Middle C for 1 beat", "clear screen", "stop all sounds"

Build it / Step by step

- 1 In "on shake", place "show icon" Angry.
- 2 Add three "play tone Middle C for 1 beat" blocks under it - that is your siren.



Real MakeCode blocks / example from these steps

Make it yours.

Backpack guard

- 3 In "on button A pressed", place "clear screen" and "stop all sounds". That resets your alarm.
- 4 Download it, rest it on your bag, arm it, and ask a friend to touch the bag.
- 5 Talk about false alarms: a bump from the table should not mean a thief. What would you change to make the alarm smarter?

Make it yours / Challenge

Make the warning sign your bag deserves: Protected by Circuit Kids, in your best sign-writing letters, on bright card. A good alarm looks serious before it makes a sound.

SKETCH YOUR DESIGN



What you learned

Sensors have sensitivity, alarms live or die by their threshold, and every false alarm is a design lesson, not a failure.

My idea / what I would improve

Which way home?

Your story. Your code. Your community.

A visitor climbs off the taxi and asks for your street. "Go north past the rank, then east at the spaza." Build a compass so your directions are always true.

Learning objective

Press A and the board shows your compass heading in degrees - then upgrade it to say N, E, S or W.

What you need

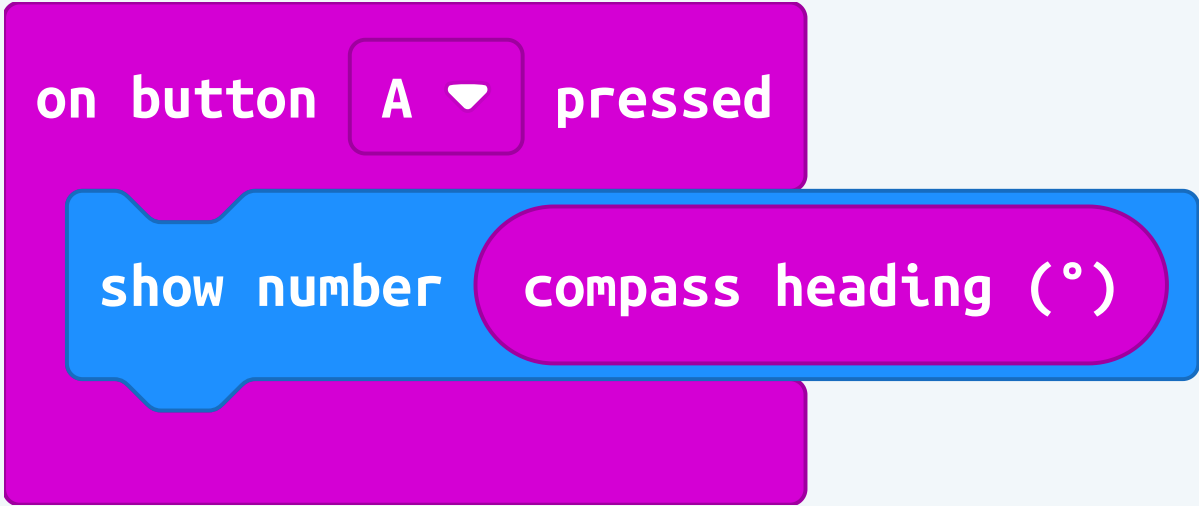
GO kit, one computer, a safe spot outside

Blocks you will use

"on button A pressed", "compass heading (°)", "show number", "if ... then ... else", "show string"

Build it / Step by step

- 1 In "on button A pressed", place "show number" with "compass heading (°)" inside it.
- 2 The first time it runs, the board asks you to calibrate: tilt it slowly until the circle of lights fills up.



```

on button A pressed
  show number compass heading (°)
  
```

Real MakeCode blocks / example from these steps

Make it yours.

Which way home?

- 3 Remember the code: 0° is north, 90° is east, 180° is south, 270° is west.
- 4 Upgrade: use "if compass heading (°) < 45 then" "show string" N, add "else if" bands for E, S and W.
- 5 Go outside with a facilitator, find north, and test it. Now give a visitor real directions home.

Make it yours / Challenge

Draw a compass-rose map of your kasi with real landmarks: the school, the rank, the spaza, the church, your home - each in its true direction. Art and geography in one frame.

MY TEST RECORD

Test / place	What happened?



Upgrade example: one way to complete the compass bands

What you learned

Calibration teaches a sensor about the real world, and conditionals turn one raw number into useful words.

My idea / what I would improve

Goalkeeper reflexes

Your story. Your code. Your community.

Street soccer final, and their striker is fast. Goalkeepers train reflexes every practice. Now you have a reaction timer to train yours.

Learning objective

The board waits a secret random time, flashes a heart - and measures how many milliseconds you take to press A.

What you need

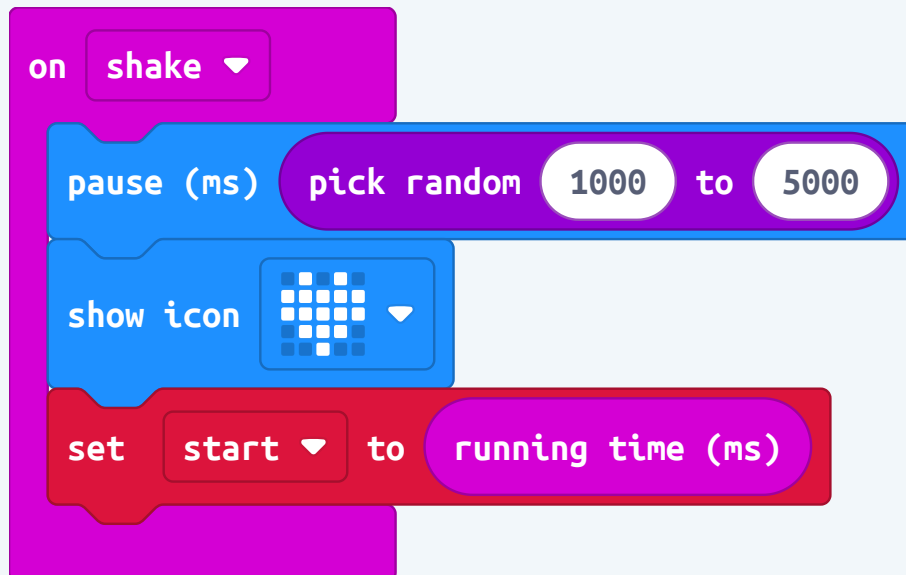
GO kit, one computer, a partner

Blocks you will use

"on shake", "on button A pressed", "pause (ms)", "pick random 1000 to 5000", "show icon", "set start to", "running time (ms)", "set time to", subtraction, "show number"

Build it / Step by step

- 1 Make two variables: start and time.
- 2 In "on shake": place "pause (ms)" with "pick random 1000 to 5000" inside, then "show icon" Heart, then "set start to" with "running time (ms)" inside.



Real MakeCode blocks / example from these steps

Make it yours.

Goalkeeper reflexes

- 3 In "on button A pressed": place "set time to" with a maths block "running time (ms) - start", then "show number" time.
- 4 Shake, stare at the board, and press A the instant the heart appears.
- 5 Under 300 milliseconds? Tell the coach. Over 500? More practice. Trade places and keep a ladder of scores.

Make it yours / Challenge

Build the club leaderboard ladder on cardboard, painted and decorated, with everyone's best time. Design a medal for the weekly champion.

MY TEST RECORD

Test / place	What happened?



Real MakeCode blocks / second event

What you learned

Events and timing let a program measure the world - including you. Evaluation means reading the number and asking what it tells you.

My idea / what I would improve

Secret signal

Your story. Your code. Your community.

Sipho is at his gate, you are at yours. No airtime, no problem - micro:bits talk to each other over radio. Send the signal: meet at the park after school.

Learning objective

Two boards on the same radio group: press A on yours and the word PARK scrolls on his.

What you need

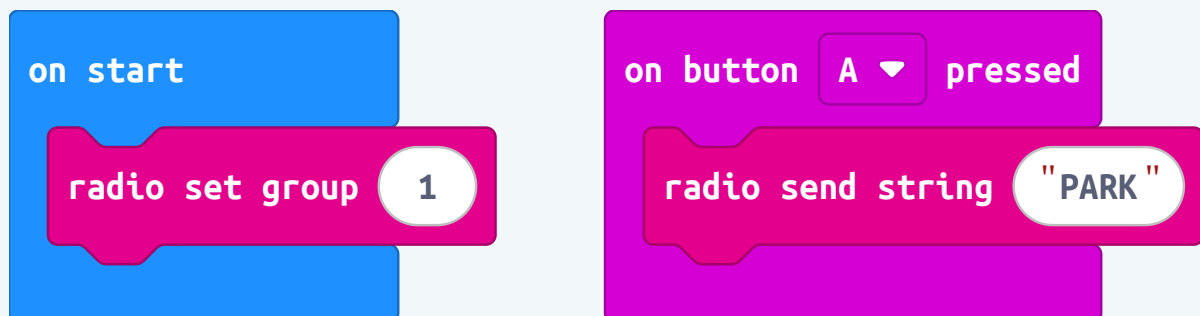
Two micro:bits (yours and a partner's), one computer

Blocks you will use

"on start", "radio set group", "on button A pressed", "radio send string", "on radio received receivedString", "show string", "play tone"

Build it / Step by step

- 1 On both boards, place "radio set group" 1 in "on start". Same group, same channel - like agreeing which corner to meet on.
- 2 On your board, in "on button A pressed", place "radio send string" PARK.



Real MakeCode blocks / example from these steps

Make it yours.

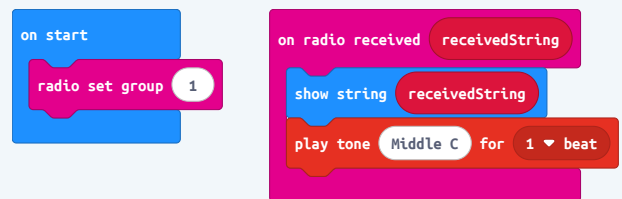
Secret signal

- 3 On his board, in "on radio received receivedString", place "show string" receivedString, and add a "play tone" so it rings.
- 4 Test across the room, then across the street. Swap roles so everyone sends and receives.
- 5 Only one board today? Pair with the facilitator's board, or load the partner .hex file from the club USB.

Make it yours / Challenge

Design your crew's secret code sheet: 1 means park, 2 means home, 3 means bring the ball. Send numbers with "radio send number" and decorate the code sheet like a real cipher.

SKETCH YOUR DESIGN



Real MakeCode blocks / second event

What you learned

Networks run on protocols - shared rules like the group number. The radio group selects matching radios; it does not encrypt messages or keep them private.

My idea / what I would improve

Club quiz buzzer

Your story. Your code. Your community.

Youth club quiz night, and every round ends the same way: "I pressed first!" Build a buzzer that remembers a press until the quizmaster resets it.

Learning objective

On one board, the first press shows your team icon and locks later presses on that board out. The quizmaster presses A+B to reset for the next question. Separate boards do not lock each other out.

What you need

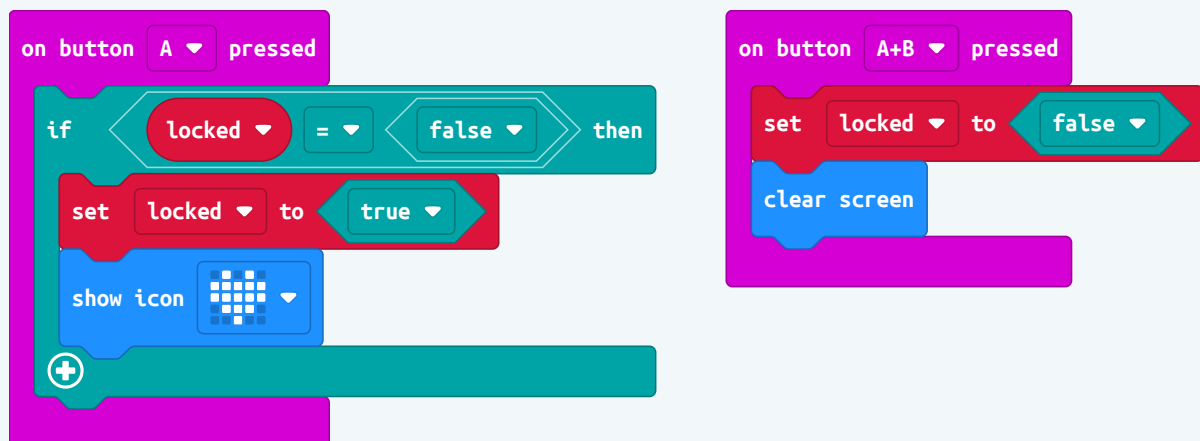
GO kit (a second board per team if you have them), one computer

Blocks you will use

"on start", "on button A pressed", "set locked to", "if ... then", "on button A+B pressed", "show icon", "clear screen"

Build it / Step by step

- 1 Make a variable called locked. In "on start", place "set locked to" false.
- 2 In "on button A pressed", place "if locked = false then". Inside: "set locked to" true, then "show icon" your team's icon.



Real MakeCode blocks / example from these steps

Make it yours.

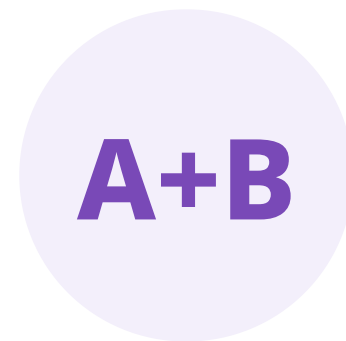
Club quiz buzzer

- 3 The lock is the fairness: once locked is true, later presses change nothing.
- 4 In "on button A+B pressed" - the quizmaster's reset - place "set locked to" false and "clear screen".
- 5 Two teams, two boards, same code, different icons. Each board locks only its own presses, so the quizmaster must still judge which team pressed first. One board only? Play against the clock instead: beat your own best reaction from Project 12.

Make it yours / Challenge

Design the quiz trophy and the season scoreboard poster for your club wall. Big letters, team colours, space for next term's champions.

SKETCH YOUR DESIGN



What you learned

State - like locked or unlocked - is what a program remembers between events, and a well-designed system makes fairness automatic.

My idea / what I would improve

Foil door alarm

Your story. Your code. Your community.

Gogo's spaza stock room needs a guard at night. You will build one from your board, two crocodile clips and a piece of kitchen foil.

Learning objective

A touch alarm: grab the foil-wrapped handle and the alarm sounds - the circuit runs through you.

What you need

GO kit, 2 crocodile leads, kitchen foil (extension tray)

Blocks you will use

"on pin P0 pressed", "on button A pressed", "show icon", "play tone", "stop all sounds", "clear screen"

Build it / Step by step

- 1 Clip one crocodile lead to pin P0 and one to GND.
- 2 Wrap the P0 end around a foil pad shaped like a door handle, and fix the GND end to a second foil pad right beside it.



Real MakeCode blocks / example from these steps

Make it yours.

Foil door alarm

- 3 In "on pin P0 pressed", place "show icon" Surprised and "play tone Middle C for 1 beat" three times.
- 4 In "on button A pressed", place "stop all sounds" and "clear screen" to reset.
- 5 Test it: touch both pads at once - your body closes the circuit and the alarm fires. Mount it on a cardboard "stock room" door and dare a friend to open it quietly.

Make it yours / Challenge

Shape the foil into something with kasi style - a snake, a gate, a burglar bar pattern - and paint the alarm box. Security can be beautiful.

SKETCH YOUR DESIGN



P0

What you learned

A circuit must be a complete loop to work, conductors like foil (and people) can close the loop, and pins turn that into an input.

My idea / what I would improve

Robot at the intersection

Your story. Your code. Your community.

The robot by the rank never sleeps: green, amber, red, keeping taxis, scholars and gogos with shopping bags moving safely. Build your own for a cardboard kasi.

Learning objective

Three LEDs - red, amber, green - wired to pins 0, 1 and 2, flashing in the real traffic-light sequence.

What you need

GO kit, 3 LEDs (red, amber, green), 3 resistors, crocodile leads, recycled card (extension tray)

Blocks you will use

"forever", "digital write pin" (P0, P1, P2; 1 and 0), "pause (ms)"

Build it / Step by step

- 1 Power off first. Wire each LED's long leg through a resistor to its pin: red to P0, amber to P1, green to P2. All three short legs go to GND. Ask your facilitator to check before powering on.
- 2 In "forever": "digital write pin P2 to 1" - green means go - then "pause (ms)" 3000, then "digital write pin P2 to 0".



Real MakeCode blocks / example from these steps

Make it yours.

Robot at the intersection

- 3 Add amber: "digital write pin P1 to 1", "pause (ms)" 1000, "digital write pin P1 to 0". Amber means get ready to stop.
- 4 Add red: "digital write pin P0 to 1", "pause (ms)" 3000, "digital write pin P0 to 0". Red means stop.
- 5 Download and watch your robot run its full cycle: green, amber, red, just like the one at the rank.

Make it yours / Challenge

Build the whole intersection from recycled boxes: the spaza, the houses, a painted taxi, and pedestrians who cross only when vehicles are stopped and an adult says it is safe. This is the A in STEAM - your kasi, in cardboard.

SKETCH YOUR DESIGN



What you learned

Digital outputs switch real lights on and off, and the same three lights in the right order - a sequence - keep a whole kasi moving.

My idea / what I would improve

Kasi beat machine

Your story. Your code. Your community.

Gqom from a passing taxi, a pennywhistle on the corner, drums at the wedding two streets down. The kasi makes music every day. Your board can too.

Learning objective

Code your own eight-note melody on button A, and a repeating beat loop on button B. Take turns playing the melody and the beat.

What you need

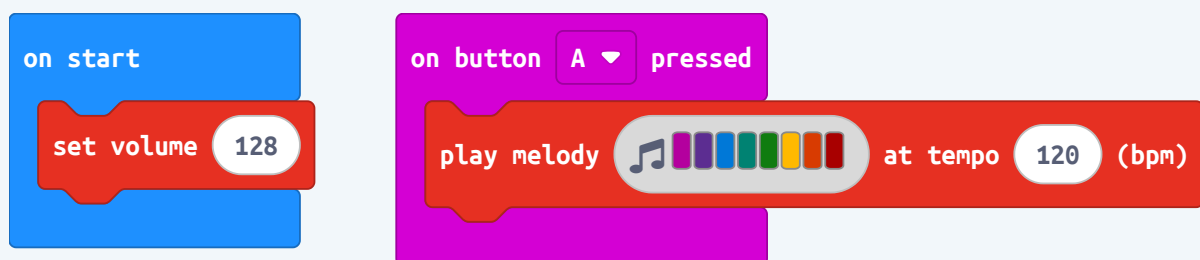
GO kit (V2 board for the speaker), one computer, headphones if the club has them

Blocks you will use

"on start", "set volume", "on button A pressed", "play melody ... at tempo 120 (bpm)", "on button B pressed", "repeat", "play tone Middle C for 1/2 beat", "play tone Low C for 1/2 beat"

Build it / Step by step

- 1 In "on start", place "set volume" 128 so everyone can hear.
- 2 In "on button A pressed", place "play melody" - tap the piano grid and write eight notes that feel like your street.



Real MakeCode blocks / example from these steps

Make it yours.

Kasi beat machine

- 3 In "on button B pressed", build a beat loop with "repeat" 4 times: "play tone Middle C for 1/2 beat", then "play tone Low C for 1/2 beat".
- 4 Download and press B: that is your drum. Wait for the beat to finish, then press A: that is your song. Take turns between beat and melody.
- 5 Perform it for the club. Loop it, speed it up, slow it down - you are the producer.

Make it yours / Challenge

Every beat needs an album cover. Name your artist self and design the cover on card - bold colours, big letters, kasi style. Music is STEAM with sound on.

SKETCH YOUR DESIGN



Real MakeCode blocks / second event

What you learned

A loop is a pattern that repeats, and a beat is exactly that. Composition and code are the same kind of thinking.

My idea / what I would improve

Gogo's special-of-the-day pointer

Your story. Your code. Your community.

Gogo's chalkboard special keeps blowing over in the wind. Build a pointer that turns to today's special at the press of a button.

Learning objective

A micro servo turns a paper arrow to point at a price on a hand-drawn dial.

What you need

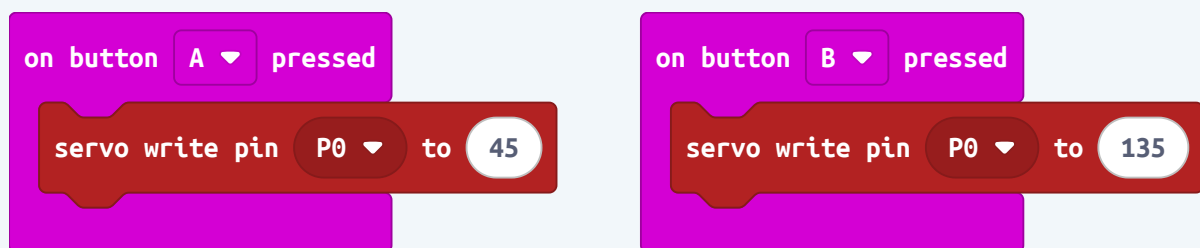
GO kit, micro servo, crocodile leads, card and a split pin (extension tray)

Blocks you will use

"on button A pressed", "servo write pin P0 to 45", "on button B pressed", "servo write pin P0 to 135"

Build it / Step by step

- 1 Safety first: the servo is powered from the battery pack, never from the board's 3V pin. Connect the servo signal wire to P0, its power wire to the battery pack, and join the GNDs together. Your facilitator checks this before anything switches on.
- 2 In "on button A pressed", place "servo write pin P0 to 45".



Real MakeCode blocks / example from these steps

Make it yours.

Gogo's special-of-the-day pointer

- 3 In "on button B pressed", place "servo write pin P0 to 135".
- 4 Press A and B and watch the servo horn swing between the two angles.
- 5 Stick a paper arrow on the horn and draw the dial on card: 45° is amagwinya, 90° is sweets, 135° is airtime. Today's special, one press away.

Make it yours / Challenge

Design the dial face in your best spaza sign-writing style - prices, colours, maybe a painted smile. A sign people love is a sign people read.

SKETCH YOUR DESIGN



45°

What you learned

Actuation is code moving the physical world, angles are data the motor understands, and mechanisms turn numbers into motion.

My idea / what I would improve

Water the spinach

Your story. Your code. Your community.

Gogo's spinach and morogo feed the family, but in this heat the soil dries out fast. Build a reminder that says "water me" before the leaves droop.

Learning objective

Two foil strips in the soil act as a probe: the board reads the moisture and shows a happy or sad face.

What you need

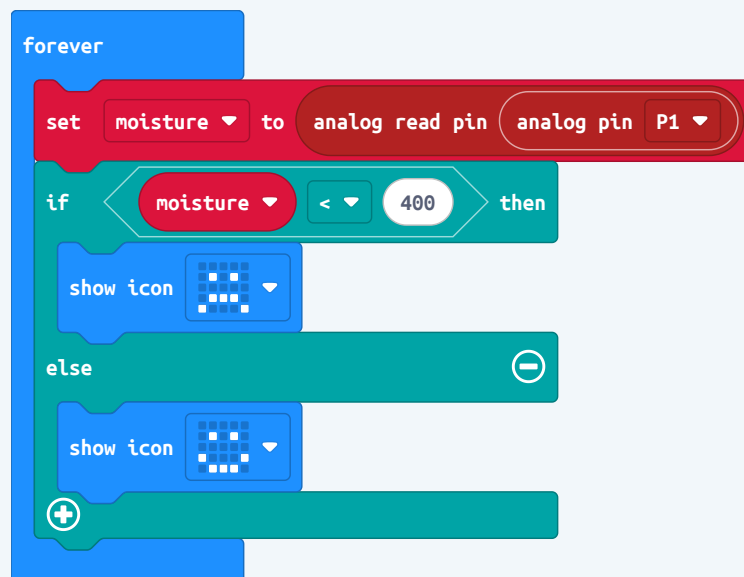
GO kit, crocodile leads, foil, a plant pot or garden bed (extension tray)

Blocks you will use

"forever", "analog read pin P1", "set moisture to", "if ... then ... else", "show icon"

Build it / Step by step

- 1 Tape two foil strips a finger-width apart on a stick. Clip 3V to one strip and P1 to the other - that is your probe.
- 2 In "forever", place "set moisture to" with "analog read pin P1" inside it (make the variable moisture first).



Real MakeCode blocks / example from these steps

Make it yours.

Water the spinach

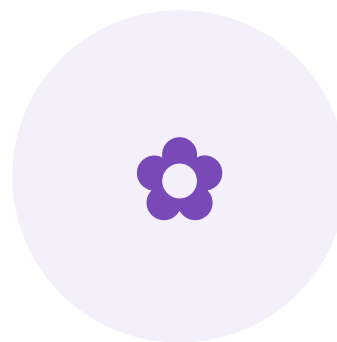
- 3 Under it, place "if moisture < 400 then" "show icon" Sad, "else" "show icon" Happy.
- 4 Test in dry soil, then just-watered soil, and write down both numbers. Now choose your own threshold - 400 is only a starting guess.
- 5 Talk about limits: foil corrodes and this probe is an experiment, not a shop-bought sensor. Real garden kits use proper probes. Knowing the limit is good engineering.

Make it yours / Challenge

Paint the marker stake, name the plant, and decorate the pot. Gogo's garden deserves a gadget that looks like it belongs there.

MY TEST RECORD

Test / place	What happened?



What you learned

Analog data comes in ranges, thresholds turn a reading into a decision, and responsible design means being honest about what a prototype can and cannot do.

My idea / what I would improve

Fix one thing in our kasi

Your story. Your code. Your community.

You have coded lights, sound, sensors, radio and movement. Every inventor who ever lived started exactly where you are standing. Find one real problem in our kasi - and fix it.

Learning objective

Your own invention: choose a problem, plan it, build it from everything you have learned, test it, and pitch it at the community showcase.

What you need

GO kit, anything from the extension tray, recycled materials, your whole imagination

Blocks you will use

Everything from Projects 1 to 19

Build it / Step by step

- 1 Choose a problem you have actually seen: a load-shedding night light, a queue counter for the clinic, a medicine reminder for gogo, a water alarm for the tank.
- 2 Plan it: sketch the invention and write down its inputs and outputs. What goes in? What comes out?

PROBLEM



PLAN



BUILD



TEST



IMPROVE

Make it yours.

Fix one thing in our kasi

- 3 Build it with any blocks and materials from the earlier projects. Borrow ideas - that is what engineers do.
- 4 Test it: fill in the evidence table - what you expected, what actually happened - and fix at least one bug you find.
- 5 Pitch it at the showcase: two minutes. The problem, your build, and one real test result. Stand tall - you made something the kasi has never seen.

Make it yours / Challenge

This one is all yours: the name, the poster, the pitch. Make it beautiful, make it proud, make it kasi.

MY EVIDENCE TABLE

I expected	It actually did



What you learned

The engineering design cycle - problem, plan, build, test, improve - plus the courage to stand up and present. That is the whole pathway in one project.

My idea / what I would improve

Keep creating.

From Grade R to Grade 7

This book is one step on a journey that runs from Grade R to Grade 7.

Circuit Kids is a South African STEAM programme - science, technology, engineering, art and maths - built offline-first for real classrooms, with one founding principle: every language is a doorway into computational thinking.

Grade R to 3: Cooper the robot

Screen-free coding in the Foundation Phase - body movement, command cards and floor mats build sequences and systems thinking before any screen appears.

Grades 4 to 7, Pack 1: this book

Twenty micro:bit projects with the GO kit, from first icon to community invention.

Grades 4 to 7, Pack 2: MakeCode Arcade

A second beginner book and arcade pack: learners design their own games and play them on a handheld console.

Circuit Kids Club

Saturday classes where buying the pack is the enrolment and monthly fees stay minimal. Learners own their packs; the club owns the laptops, expansion kits and specialist equipment.

Build the next possibility.

The children who will hold it next

Circuit Kids is being built to last: a non-profit programme with registration in progress, a full curriculum, a tested kit and books like this one. The print edition carries a designed acknowledgement page and back-cover panel for the sponsors and partners who fund it - every logo with written permission, and editorial control always with Circuit Kids.

What you are holding is proof of determination: a finished learning resource, ready for pilots and workshops, made for the children who will hold it next.



Acknowledgements

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**Every language is a doorway
into computational thinking.**

Made for Circuit Kids.

Township edition / Pilot and workshop resource

Circuit Kids is an independent programme. This book is not endorsed by, or affiliated with, the Micro:bit Educational Foundation or Microsoft. BBC micro:bit is a trademark of the Micro:bit Educational Foundation; Microsoft MakeCode block names and screenshots belong to Microsoft. Cover art in this document is illustrative. All lessons, stories and code were written originally for Circuit Kids, with reputable computing-education libraries consulted as reference only.

Visual treatment

Original illustrative township scenes; original simplified micro:bit schematic; authentic block graphics rendered with Microsoft MakeCode. Illustrations are not product photographs.

Edition note

Township Edition - October 2026. Visual edition for pilots and sponsors. Registration in progress.

Reference sources

micro:bit features: microbit.org/get-started/features/overview/

MakeCode block rendering: makecode.microbit.org/blocks-embed

Code. Build. Create.
Solve. Grow.